

Introduction to Prompt Engineering



Prompt engineering is a crucial concept in the realm of AI and technology. It involves crafting inputs that guide AI models to produce desired outputs, significantly enhancing interactions and results.

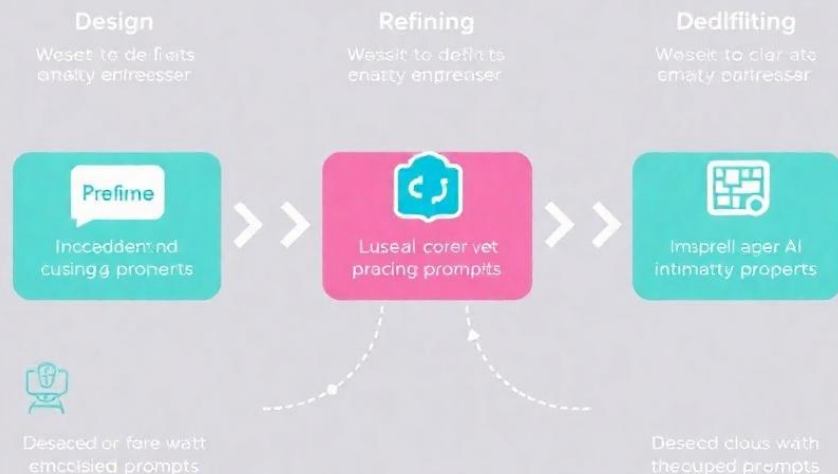


What is Prompt Engineering

Prompt engineering is the process of designing and refining prompts to optimize AI responses. Its importance lies in improving the quality of AI interactions, making them more relevant and effective for users.

Prompt Engineering

Overall, your prompts fit the scenarios and the resulting output is fast, accurate, and useful. This is the power of prompt engineering.



Designing is a quick and easy process that can be done by anyone. The AI interface is designed to be easy to use and to provide a good user experience.

Real-World Applications



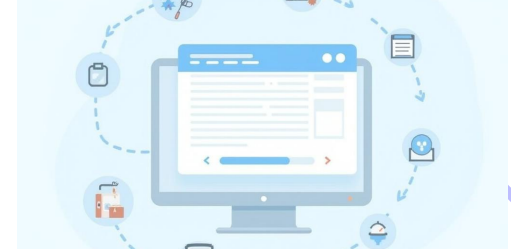
Chatbots

AI-driven chatbots provide curriculum support, assessment, Quizzes, Student engagement, etc..



Risk Analysis

Prompt engineering aids in analyzing curriculum by generating insights based on data inputs.



Report Automation

Automating report generation through prompts saves time and ensures accuracy in education..

Examples of AI Tools Using Prompts

ChatGPT

A conversational AI tool that generates human-like text based on user prompts, useful for various applications.

Claude

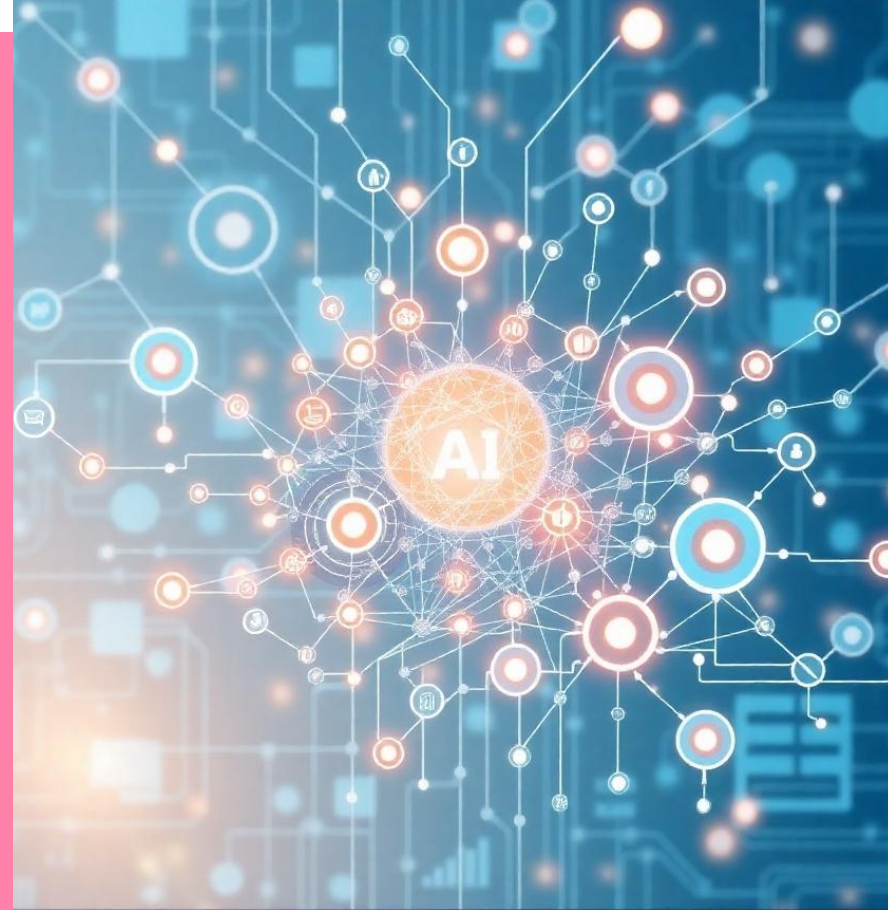
An AI model designed for generating detailed reports and analyses, leveraging prompt engineering for accuracy.

Gemini

A tool that creates narratives and content based on prompts, enhancing creative writing and storytelling.

Understanding AI Models

AI models interpret prompts through complex algorithms that analyze input data. Understanding this process is essential for crafting effective prompts that yield accurate and relevant responses.



Basic Prompt Structure

Clear Objective

Define what you want the AI to achieve.

Context

Provide necessary background information.

Specific Instructions

Detail any specific requirements for the response.

How do AI prompts work?



Asking a query

The query is analyzed by natural language understanding (NLU), a subtask of NLP.



Processing the query

The AI model scans the available data to search for relevant information by using sub-tasks such as named entity recognition (NER) and intent recognition.



Answering the query

This part uses natural language generation (NLG), where different data points, based on the question string in the prompt, are combined to generate an answer.

Types of Prompts

- **Instructive Prompts:** Guide the AI to perform specific tasks or follow particular instructions.
- **Descriptive Prompts:** Used to elicit detailed information or explanations.
- **Conversational Prompts:** Facilitate dialogue and interaction, making the AI more engaging and user-friendly.

Role-Based Prompts

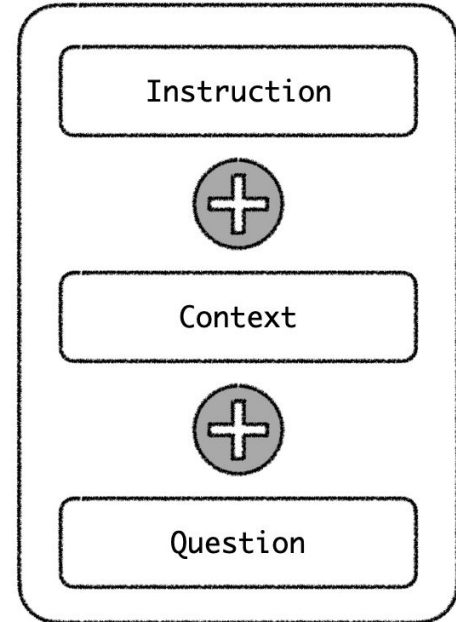
Role-based prompts assign a specific role to the AI, enhancing its focus. For example, "Act as a History Teacher" directs the AI to provide insights from a financial perspective.



Contextual Prompts

Contextual prompts provide specific scenarios for the AI to analyze. For instance, "Given a class of 30 mixed-ability learners in SS1" helps the AI understand the context and deliver targeted insights.

Contextual Prompt Structure





Best Practices in Prompt Engineering

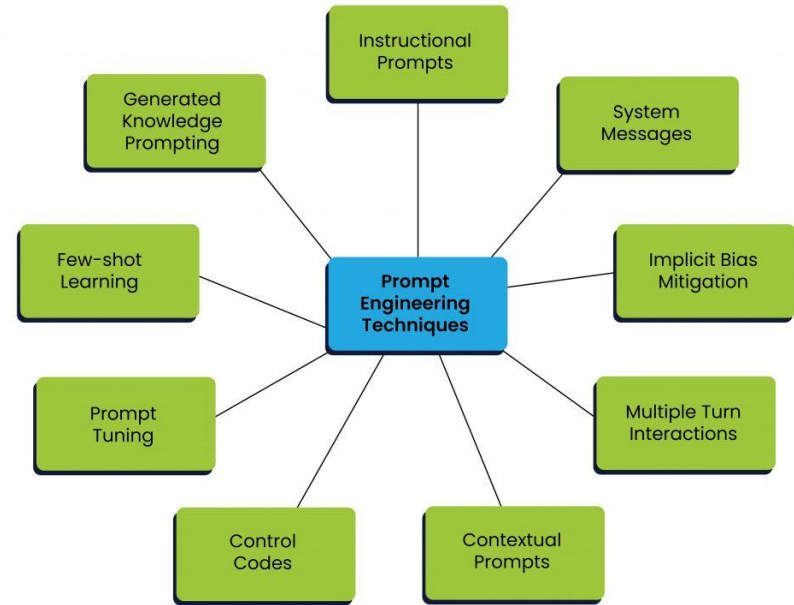
Simplicity and Clarity: Keep prompts straightforward and easy to understand.

- Avoid unnecessary complexity to enhance AI performance.
- Iterative Testing: Continuously test and refine prompts based on the AI's responses to achieve optimal results.

Common Mistakes in Prompt Engineering

Over-Engineering Prompts: Avoid making prompts overly complex, which can confuse the AI and lead to irrelevant outputs.

- Lack of Clarity: Ensure prompts are clear and specific to prevent misunderstandings and improve response quality.



Ensuring Compliance and Data Security

When using AI in finance, compliance with regulations and data security is paramount. Prompt engineering can help ensure that sensitive information is handled appropriately and securely.



Bias in AI Decision-Making

It's crucial to address potential biases in AI outputs. Prompt engineering should consider fairness and inclusivity to mitigate bias in decision-making processes.



Conclusion and Q&A

In conclusion, prompt engineering is a vital skill for enhancing AI interactions, particularly in finance. By understanding its principles and best practices, you can effectively leverage AI tools.